

Math 1C Test 2
Fall 2025

Problems are 10 points each. Please show work to get credit.

1. Given $z = f(x, y) = x^3 - 2xy$, find (leave solutions in math expression form):

a) $f(1, 3)$ and $f(1.01, 2.98)$ and Δz .

b) use total differential dz to approximate Δz .

2. Find the limit:

a)
$$\lim_{(x, y) \rightarrow (0, 0)} \frac{x + y}{x^2 + y^2}$$

b)
$$\lim_{(x, y) \rightarrow (0, 0)} \frac{x^2}{\sqrt{x^2 + y^2}}$$

3. Given the function $f(x, y) = \frac{1}{4}y^2 - x^2$, find the directional derivative of the function at (1,4) in the direction of $\mathbf{v} = 2\mathbf{i} + \mathbf{j}$.

4. Given the function $g(x, y) = \ln\sqrt{xy}$, find the maximum value of the derivative of the function at (2, 1).

5. Examine the function $f(x, y) = yx^2 - 2x^2 - y^2$ for local extrema and saddle points.

6. Given the function $f(x, y) = x^2 y^3$:

- a) Find the directional derivatives from **P(1,1)** to **Q(4,2)** at point P.
- b) In what direction does the rate of change increase the most? What is the maximum rate of increase?
- c) In what direction does the rate of change decrease the most? What is the maximum rate of decrease?
- d) Find the equation of the tangent line at P.

7. Examine the **absolute extrema** for the function $f(x, y) = x^2 - 2xy + y^2$ in the region R. R is a triangle with vertexes at (0,0), (2,0), and (0,3). Be sure to graph R.

8. A rectangular box has a volume of 36 ft^3 . The bottom of the box can be made from a material that costs $\$1/\text{ft}^2$, but the material for the top and sides costs $\$8/\text{ft}^2$. Use Lagrange multipliers to find the **dimensions** for the box so the corresponding **cost** of the box is minimum.